FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2025

MATHEMATICS

Past Paper 2025 • Paper Code: 1HA-I 25008 — 30081 (Version B) / 32081 (Version D)

Retyped faithfully from the original board paper — same questions, no changes.

VERSION B

MATHEMATICS HSSC-I — SECTION – A (Objective Type)

Time Allowed: 25 Minutes | **Maximum Marks:** 20 | **Paper Code:** 30081 (Version B)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. What are the multiplicative factors of z^2+4z+5?

(A) $(z+2-i)(z+2+i)$ (B) $(z-2-i)(z-2+i)$

(C) $(z+2-2i)(z+2+2i)$ (D) $(z-2-2i)(z-2+2i)$

Q2. The principal argument(θ) of the complex number $z=-1+i\sqrt{3}$ is:

(A) $-\pi/3$ (B) $\pi/3$

(C) $2\pi/3$ (D) $5\pi/3$

Q3. A system of linear equations is said to be inconsistent if it has:

(A) No solution (B) At least one solution

(C) Infinitely many solutions (D) Exactly two solutions

Q4. If A is a 3 x 3 matrix, then |3A| results

(A) 3|A| (B) 6|A|

(C) 9|A| (D) 27|A|

Q5. In an arithmetic sequence, $a_1=5$ and $a_{15}=75$. What is the common difference?

(A) 3 (B) 4

(C) 5 (D) 6

Q6. If 1/2, 1/x, 1/5 are in harmonic sequence then what is the value of x?

(A) 2/7 (B) 7/2

(C) 7/10 (D) 10/7

Q7. What is the first term of an infinite geometric series whose sum is 10, and common ratio 0.6?

(A) 4 (B) 5

(C) 6 (D) 7

Q8. What is the sum of the series up to 4 terms? $1+3+5+\cdots+(2n-1)=n^2\ \forall n\in N$

(A) 7 (B) 9

(C) 15 (D) 16

Q9. What is the middle term in the expansion of $(x+y)^6$?

(A) (6 choose 2)x^3y^3 (B) (6 choose 3)x^3y^3

(C) (6 choose 3)x^4y^2 (D) (6 choose 2)x^4y^2

Q10. In signal processing, P(x) represents a signal. If $P(-1)=0$, then the signal has:

(A) A zero at $x=-1$ (B) Unstablity

(C) Periodicity (D) Error

Q11. If dot product of two vectors is zero then angle between them is:

(A) 180° (B) 90°

(C) 45° (D) 0°

Q12. If $\bar{a}.\bar{b}\times\bar{c}=12$ then volume of parallelepiped with $\bar{a}$, $\bar{b}$ and $\bar{c}$ as co–terminal edges is:

(A) 6 (B) 12

(C) 24 (D) 36

Q13. A force $\bar{F}=2\hat{i}+\hat{j}+3\hat{k}$ moves an object along a displacement vector $\bar{d}=3\hat{i}+4\hat{j}-2\hat{k}$. What is the work done?

(A) −4 (B) 10

(C) 4 (D) 16

Q14. If $\tan\theta=2$, what is the value of $\cos\theta$ when θ is in the first quadrant?

(A) $1/\sqrt{3}$ (B) $2/\sqrt{5}$

(C) $\sqrt{5}$ (D) $1/\sqrt{5}$

Q15. If $\sin x=3/5$, what is the value of $\sin 2x$? $(0°<x<90°)$

(A) 12/25 (B) 24/25

(C) 7/25 (D) 25/25

Q16. If graph of a function is symmetric about origin (e.g. Sin x), what type of function is it?

(A) Even (B) Odd

(C) Neither Even nor Odd (D) Periodic but not Symmetric

Q17. The range of a trigonometric function $4\sin 4x$ is:

(A) $[-1,1]$ (B) $[-4,4]$

(C) $[-16,16]$ (D) $(-\infty,+\infty)$

Q18. The expression $\sin 75°\cos 15°$ is equivalent to:

(A) $\frac{1}{2}(\sin 90°+\sin 60°)$ (B) $\frac{1}{2}(\cos 90°+\cos 60°)$

(C) $\frac{1}{2}(\sin 60°-\sin 90°)$ (D) $\frac{1}{2}(\cos 60°-\cos 90°)$

Q19. How many 5-letter words can be formed from the word 'MATHS' without repeating any letter?

(A) 20 (B) 60

(C) 120 (D) 240

Q20. In how many different ways a player can select 6 numbers from a set of 49 numbers, in a lottery?

(A) ${}^{49}C_6$ (B) 49!/6!

(C) ${}^{49}P_6$ (D) 6!

> Note: Important Formula: $S_\infty = a_1/(1-r)$

SECTIONS – B & C (Subjective Type)

Time Allowed: 2:35 Hours | **Maximum Marks:** 80 | **Paper Code:** 1HA-I 25008 (B)

Q2. (Marks (12 x 4 = 48)) Q.2 Solve the following parts.

(i) [04] Solve the quadratic equation $3z^2 + 2z + 2 = 0$, $z \in C$ by completing the square method. — OR (04) — A triangular park has its three corners located at points A (2, 3, 1), B (5, 7, 2), and C (1, 4, 3). Find area of the park using cross product of vectors.

(ii) [04] Verify that: $\sin\theta \cdot \sin(\pi/3-\theta) \cdot \sin(\pi/3+\theta) = \frac{1}{4}\sin 3\theta$ — OR (04) — Verify that: (n choose r) + (n choose r−1) = (n+1 choose r)

(iii) [04] Find the value of p such that vectors $\hat{i} + 2\hat{j} + p\hat{k}$, $3\hat{i} + p\hat{j} + 4\hat{k}$ and $2\hat{i} + 3\hat{j} + 4\hat{k}$ are coplanar. — OR (04) — Find the maximum and minimum values of the function $f(\theta) = 1/(3+5\cos(2\theta+\pi))$

(iv) [04] If $P(x) = x^4 - 6x^3 + 11x^2 - 6x$, then (a) Divide P(x) by (x−1) using synthetic division (b) Solve the resulting depressed equation. — OR (04) — A harmonic sequence has 2nd term 1/6 and 4th term 1/12. Find general term of the sequence.

(v) [04] Without drawing graph find range, amplitude, period and frequency of the function $y = -4\cos(7x-\pi)$ — OR (04) — Find the term containing x^3 in the expansion of $(x + 2)^6$.

(vi) [04] In an arithmetic sequence, the 5th term is 15, and the 12th term is 50. Find a) Common difference and b) First term of the sequence. — OR (04) — Verify that: $(1+\sin 2\theta+\cos 2\theta)/(1+\sin 2\theta-\cos 2\theta) = \cot\theta$

(vii) [04] Verify that $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (b-a)(c-a)(c-b)$ — OR (04) — If $\tan\theta = 3/4$ with $\pi < \theta < 3\pi/2$, find the exact values of (a) $\sin(\theta/2)$ and (b) $\cos(\theta/2)$ without using calculator.

(viii) [04] A bouncing ball rebounds 80% of its previous height. If the ball is dropped from a height of 50 meters. Find the total distance traveled by the ball before coming to rest. — OR (04) — How many words can be formed by using the letters from the word "EDUCATION" such that all the vowels are never together?

(ix) [04] Use the principle of mathematical induction to prove that $1^3 + 2^3 + 3^3 + \ldots + n^3 = ((n(n+1))/2)^2$ $\forall n \in N, n \geq 1$. — OR (04) — Find rank of the matrix $\begin{bmatrix} 1 & 1 & 0 & -2 \\ 2 & 0 & 2 & 2 \\ 4 & 1 & 3 & 1 \end{bmatrix}$

(x) [04] If $z_1 = 10(\cos 100° + i\sin 100°)$ and $z_2 = 5(\cos 40° + i\sin 40°)$, then find the following in polar form: (a) $z_1 \cdot z_2$, (b) z_1/z_2. — OR (04) — A solar panel's sunlight reception is modeled as $P(\theta) = 100\cos(3\theta-90°)$, with $0° \leq \theta \leq 180°$ (a) Find angle for the maximum sunlight and the max sunlight received. (b) Find angle for the minimum sunlight and the min sunlight received.

(xi) [04] A force $\bar{F} = 6\hat{i} + 8\hat{j} + 4\hat{k}$ acts on an object. The object moves from A (1,2,3) to point B(4,6,5). (a) Find the displacement vector $\bar{d}$. (b) Calculate work done by the force $\bar{F}$. — OR (04) — If $P(x) = ax^3 + bx^2 + 2x - 1$, leaves remainder 5 when divided by (x−1) and leaves remainder 3 when divided by (x+1). Find values of a and b using remainder theorem.

(xii) [04] Solve x + y + z = 3; y + z = 2; z = 2 using Cramer's rule: — OR (04) — Use binomial theorem to approximate the value of $\sqrt{101} \times \sqrt{99}$ up to three places of decimal.

> Note: SECTION – C (Marks 32) — Note: Solve the following Questions. (Use of the graph paper is not necessary. Candidates can makes their own grid on answer book) (4 x 8 = 32)

Q3. (Marks 08) Q.3 If $P(x) = 2x^3 + x^2 - 13x + 6$, then a) Verify that (x + 3) is a factor of P(x). b) Factorize P(x) using synthetic division. c) Find all the roots of P(x). d) Write complete factorized form of P(x) and confirm by expanding. — OR (08) — If $\bar{a}$

= 3î − 2ĵ + 4k̂ and b̄ = î + 2ĵ − k̂, then (a) Find ā×b̄. (b) Find (ā×b̄)·ā and verify that ā×b̄ ⊥ ā. (c) Find (ā×b̄)·b̄ and verify that ā×b̄ ⊥ b̄

Q4. (Marks 08) Q.4 If x is very small such that its square and higher powers can be neglected, then show that: (16+4x)^(3/4) / ((4+x)√(9−6x)) ≈ 2/3 + 13x/72 — OR (08) — Verify the fundamental law of trigonometry: cos(α−β) = cosαcosβ + sinαsinβ Where α, β are real angles in standard position and α > β.

Q5. (Marks 08) Q.5 Solve x + 2y − 3z = 4, 2x − 3y + 4z = 5, 3x + 4y − 5z = 6 using the Gaussian elimination method. — OR (08) — y = 3cos2x; −π/2 ≤ x ≤ π/2 a) Make table of values for given interval b) Draw the graph of the function for given interval

Q6. (Marks 08) Q.6 For an arithmetic-geometric series: $1 + 5(1/2)^1 + 9(1/2)^2 + 13(1/2)^3 + \ldots$ (a) Find general term of the series. (b) Sum the series upto n terms, (c) Sum the series upto infinite terms. — OR (08) — A company has 7 engineers and 5 managers. In how many ways can a project team of 5 be selected if it includes: (a) 3 engineers and 2 managers? (b) 4 engineers and 1 manager? (c) At least 4 engineers? (d) At the most 1 manager?

VERSION D

MATHEMATICS HSSC-I — SECTION – A (Objective Type)

Time Allowed: 25 Minutes | **Maximum Marks:** 20 | **Paper Code:** 32081 (Version D)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. What are the multiplicative factors of $z^2 + 9$?

(A) $(z+3i)(z-3i)$ (B) $(z-3)(z+3)$

(C) $(z+3)^2$ (D) $(z-3i)^2$

Q2. The rectangular form of $4(\cos 2\pi/3 + i \sin 2\pi/3)$ is:

(A) $-2+2i\sqrt{3}$ (B) $-2-2i\sqrt{3}$

(C) $2+2i\sqrt{3}$ (D) $2-2i\sqrt{3}$

Q3. A system of linear equations is said to be inconsistent if it has:

(A) No solution
(B) At least one solution
(C) Exactly one solution
(D) Exactly two solution

Q4. If |a b c; d e f; g h i| = 5, then |2a 2b 2c; 3d 3e 3f; g h i| =

(A) 6
(B) 10
(C) 15
(D) 30

Q5. In an arithmetic series, if a=2, d=4 then S_{15}=

(A) 240
(B) 450
(C) 480
(D) 510

Q6. What is the sum of first two terms of an arithmetic-geometric sequence $T_n = (2n+1)\cdot 3^{n-1}$? (n∈N)

(A) 5
(B) 11
(C) 18
(D) 23

Q7. For what value of x geometric series $1+2x+4x^2+8x^3+...$converges? (r<1)

(A) 1/2
(B) 1/4
(C) 1
(D) 2/1

Q8. The number of ways to climb a 10-step staircase (taking either 1 or 2 steps at a time) is 89. Which method justifies the best conclusion?

(A) Empirical counting
(B) Pascal's Triangle
(C) Mathematical Induction
(D) Binomial Theorem

Q9. What is the middle term in the expansion of $(x+y)^6$?

(A) (6 choose 2)x^3y^3
(B) (6 choose 3)x^3y^3
(C) (6 choose 3)x^4y^2
(D) (6 choose 2)x^4y^2

Q10. The division of $2x^3-5x^2+3x-1$ by $(x-2)$ gives remainder:

(A) −3
(B) 0
(C) 5
(D) 1

Q11. If dot product of two vectors is zero then angle between them is:

(A) 180°
(B) 90°
(C) 45°
(D) 0°

Q12. If $\bar{a} \perp \bar{b}$, then projection of $\bar{a}$ on $\bar{b}$ is:

(A) −1
(B) 0
(C) 1
(D) Undefined

Q13. What is the volume of a parallelepiped formed by vectors $\hat{i}+\hat{j}$, $\hat{j}+\hat{k}$ and $\hat{k}+\hat{i}$?

(A) 0 (B) 1

(C) $\sqrt{2}$ (D) 2

Q14. $\sin 75° \cos 15° =$

(A) $\frac{1}{2}(\sin 90° + \sin 60°)$ (B) $\frac{1}{2}(\cos 90° + \cos 60°)$

(C) $\frac{1}{2}(\sin 60° - \sin 90°)$ (D) $\frac{1}{2}(\cos 60° - \cos 90°)$

Q15. What is value of $\cos(\theta/2)$, if $\cos\theta = 1/2$; $(0 < \theta < 90°)$

(A) $\sqrt{3}/2$ (B) $1/2$

(C) $1/\sqrt{2}$ (D) $1/(2\sqrt{2})$

Q16. Which of the following is an odd function?

(A) $f(x)=\cos x$ (B) $f(x)=\sin x+\cos x$

(C) $f(x)=\tan x$ (D) $f(x)=\sec x$

Q17. The range of $f(x)=3\cos x+4$ is:

(A) $[-1,1]$ (B) $[1,7]$

(C) $[-3,3]$ (D) $[4,7]$

Q18. Which function's graph is symmetric about y-axis?

(A) $y=\sin x$ (B) $y=\tan x$

(C) $y=\cos x$ (D) $y=\csc x$

Q19. A cell phone password consists of 4 distinct digits (0-9). How many possible passwords can be set?

(A) ${}^{10}C_4$ (B) ${}^{10}P_4$

(C) ${}^{9}C_4$ (D) ${}^{9}P_4$

Q20. How many 5-letter words can be formed from the word 'TABLE' without repeating any letter?

(A) 20 (B) 60

(C) 120 (D) 240

> Note: Important Formula: $S_n = n/2[2a+(n-1)d]$

SECTIONS – B & C (Subjective Type)

Time Allowed: 2:35 Hours | **Maximum Marks:** 80 | **Paper Code:** 1HA-I 25008 (D)

Q2. (Marks (12 x 4 = 48)) Q.2 Solve the following parts.

(i) [04] Express the following in rectangular form. a) [5(cos45° + isin45°)][4(cos15° + isin15°)] b) [27(cos120° + isin120°)] / [3(cos60° + isin60°)] — OR (04) — A mechanic applies a force $\bar{F} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ at point A (14,−3, 6) to turn a bolt with pivot at P (2,−2,1). Find: (a) the moment arm $\bar{r}$ and (b) the moment vector $\bar{M}$ about the bolt.

(ii) [04] Verify that: cosθ·cos(π/3−θ)·cos(π/3+θ) = ¼ cos3θ — OR (04) — Verify that: ${}^{n}P_r = {}^{n-1}P_r + r\cdot{}^{n-1}P_{r-1}$

(iii) [04] Find the value of λ such that vectors $\hat{i} + 4\hat{j} + 2\hat{k}$, $2\hat{i} - \hat{j} + 3\hat{k}$ and $\lambda\hat{i} + 3\hat{j} + 5\hat{k}$ are coplanar. — OR (04) — Find the maximum and minimum values of the function: f(θ) = 1/(3+4sin(5θ+π))

(iv) [04] If $P(x) = 3x^3 - 5x^2 - 16x + 12$, then (a) Use factor theorem to verify that (x + 2) is a factor of P(x). (b) Apply synthetic division to factorize P(x) completely. — OR (04) — A harmonic sequence has 3rd term 1/8 and 5th term 1/20. Find: (a) the general term and (b) the 15th term of the sequence.

(v) [04] Without drawing graph find range, amplitude, period and frequency of the function y = −3sin(5x + π/3) — OR (04) — A force $\bar{F} = 5\hat{i} + 3\hat{j} + 4\hat{k}$ is applied to move an object from A(2,3,1) to B(5,7,4). Find work done by the force.

(vi) [04] In an arithmetic sequence, first term is 12, and 10th term is 48. Find: a) the common difference and b) the sum of the first 12 terms of the sequence. — OR (04) — Verify that: (cosθ+cos4θ+cos7θ)/(sinθ+sin4θ+sin7θ) = cot4θ

(vii) [04] Verify that $|a+k\ a\ a;\ a\ a+k\ a;\ a\ a\ a+k| = k^2(3a+k)$ — OR (04) — If sinθ = −3/5 and θ lies in 3rd quadrant, find the exact values of (a) cos(180°+θ) (b) tan(270°−θ)

(viii) [04] A falcon dives from a height of 100 meters to catch its prey and then ascends back to 85% of its previous height after each unsuccessful attempt. Find the total vertical distance travelled by the falcon before stopping its attempts. — OR (04) — A boat is sailing north at 10 km/h but is also being pushed east by ocean currents at 5 km/h. Find: (a) actual speed (V) of the boat. (b) direction angle (θ) of the boat relative to due north. — [A diagram of the velocity triangle (V, V_N, V_E, θ) is printed next to the question in the source scan]

(ix) [04] Use principle of mathematical induction to prove that: $1^2 + 2^2 + 3^2 + \cdots + n^2 = n(n+1)(2n+1)/6\ \forall n \in N,\ n \geq 1$ — OR (04) — Find rank of the matrix: [1 1 0 −2; 2 0 2 2; 4 1 3 1]

(x) [04] Solve the simultaneous linear equations 2iz − 3iw = 4 and −3iz + 5iw = 9 carrying complex coefficients (i = √−1). — OR (04) — A bakery offers 2 types of breads, 7 types of fillings and 5 types of sauces for a custom sandwich: (a) How many different sandwich combinations can be made? (b) How many weeks can a customer eat a different sandwich each day without repetition?

(xi) [04] Find the term containing x^4 in the expansion of $(2x - 3)^7$. — OR (04) — A checksum polynomial $C(x) = x^4 + px^2 + qx + 6$ is used to detect errors in transmitted data. (a) If C(1)=10 and C(−1)=4, set equations for p and q. (b) Solve the equations to find values of p and q.

(xii) [04] Solve the system $x - y + z = 1$; $x - z = 2$; $z = 3$ using Cramer's rule. — OR (04) — Use binomial theorem to approximate the value of $(1.03)^4/(0.98)^3$ upto three places of the decimal.

> Note: SECTION – C (Marks 32) — Note: Solve the following Questions. (Use of the graph paper is not necessary. Candidates can makes their own grid on answer book) (4 x 8 = 32)

Q3. (Marks 08) Q.3 A polynomial $P(x) = x^4 + ax^3 - 5x^2 + bx + c$ (a) Leaves remainder 0 when divided by $(x - 0)$ (b) Leaves remainder -3 when divided by $(x - 2)$ (c) Leaves remainder -4 when divided by $(x + 1)$ Use remainder theorem to find values of a, b and c — OR (08) — If $\bar{a} = 5\hat{i} - 3\hat{j} + \hat{k}$ and $\bar{b} = -2\hat{i} + 6\hat{j} + 4\hat{k}$, then (a) Find $\bar{a}\times\bar{b}$ (b) Find $(\bar{a}\times\bar{b})\cdot\bar{a}$ and verify that $\bar{a}\times\bar{b} \perp \bar{a}$ (c) Find $(\bar{a}\times\bar{b})\cdot\bar{b}$ and verify that $\bar{a}\times\bar{b} \perp \bar{b}$

Q4. (Marks 08) Q.4 If x is very small such that its square and higher powers can be neglected, then show that (8+2x)^(2/3) / ((2+x)(√(4−2x))) ≈ 1 − x/12 — OR (08) — Verify the fundamental law of trigonometry: $\cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$ Where α, β are real angles in standard position and $\alpha > \beta$.

Q5. (Marks 08) Q.5 Solve the system $x + y + z = 6$, $2x + 3y + z = 11$, $3x + y + 2z = 8$ using the Gaussian elimination method. — OR (08) — $y = 2\sin(2x)$; $-\pi/2 \le x \le \pi/2$ a) Make table of values for given interval b) Draw the graph of the function for given interval

Q6. (Marks 08) Q.6 For an arithmetic-geometric series: 1(1) + 3(1/2) + 5(1/4) + 7(1/8) + 9(1/16) + ... (a) Find n^{th} term of the series, (b) Sum the series up to n terms and (c) Sum the series up to infinite terms — OR (08) — A basketball club has 7 forwards and 5 guards. In how many ways a team of 5 can be selected if it includes: (a) 3 forwards and 2 guards? (b) 4 forwards and 1 guard? (c) At least 4 forwards? (d) At the most 1 guard?